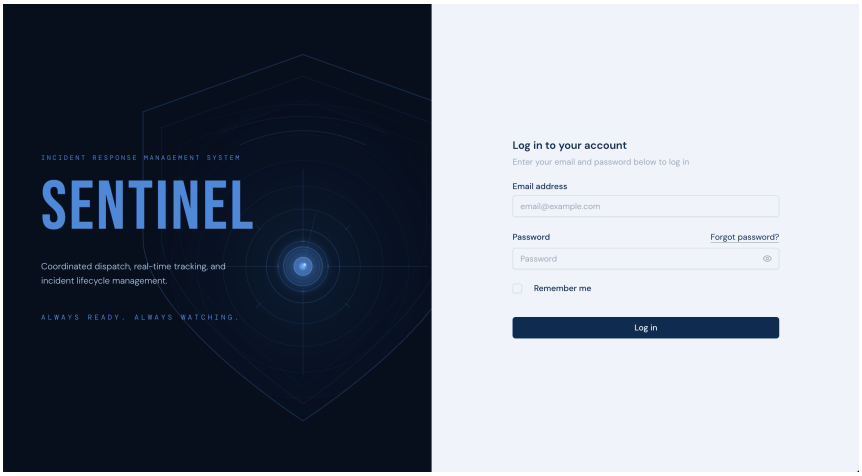
INCIDENT RESPONSE MANAGEMENT SYSTEM

SENTINEL

Sentinel is a locally-built platform for end-to-end emergency incident management. It connects CDRRMO dispatchers, field responders, and city officials through real-time situational awareness across the full incident lifecycle.

The platform consists of a web-based operations center (Intake Station + Dispatch Console), a mobile responder PWA, and a public reporting web app.



Platform Roles

REPORT	RESPOND	CONTROL
Citizen Reporter	Field Responder	Intake Operator
		Dispatch Operator
		Supervisor
		Administrator

- Zero recurring license fees — fully owned codebase
- Real-time WebSocket communication between all layers
- 5-channel incident intake: SMS, Mobile App, Voice/911, IoT Sensors, Walk-in
- Deployed for Butuan City CDRRMO, Caraga Region XIII

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System Overview

Incident Lifecycle

Report → Intake → Triage → Dispatch → Response → Resolution → Reporting

System Layers

LAYER	PRIMARY USERS	KEY FUNCTION
Intake	Operator	Receive and triage incident reports from all channels
Dispatch	Dispatcher	Assign units, track live positions, manage queue
Responder	Field Responder	Receive assignments, navigate, log scene data
Integration	System	Connect with external agencies and national systems
Analytics	Supervisor, Mayor's Office	KPI monitoring, reporting, compliance

Incident Channels

CHANNEL	SOURCE	METHOD
SMS	General public	Semaphore API gateway
Mobile App	General public	Public-facing PWA
Voice / 911	General public	Manual dispatcher entry
IoT Sensors	Infrastructure	Webhook (flood gauges, fire alarms)
Walk-in / Radio	Barangay officials	Desk entry or radio transcription

Priority Classification

PRIORITY	LEVEL	TRIGGER TYPES
P1	Critical	Cardiac Arrest, Structure Fire, Drowning, Building Collapse
P2	High	Road Accident, Assault, Stabbing, Gas Leak, Flood
P3	Medium	Domestic Violence, Seizure, Missing Person
P4	Low	Non-urgent, informational

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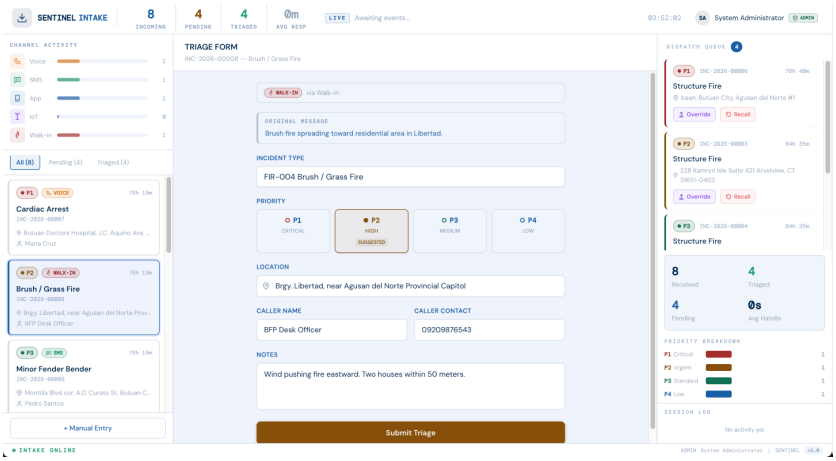
INTAKE LAYER

Intake Station

Multi-Channel Triage Hub

The Intake Station is the nerve center for receiving and processing emergency reports. Operators monitor all five incoming channels simultaneously, triage incidents using structured forms, and push validated incidents to the dispatch queue.



Live triage interface with channel monitoring, structured forms, and dispatch queue

Key Capabilities

- **Real-time channel monitoring** — SMS, App, Voice, IoT, Walk-in feeds with live counters
- **Structured triage form** — Auto-priority classification (P1 through P4) with operator override
- **Geocoding** — Automatic barangay detection via PostGIS spatial queries
- **Session metrics** — Incoming count, pending, triaged, average response time
- **Priority breakdown** — Color-coded dashboard indicators per priority level
- **Dispatch queue** — Priority-ordered queue with acknowledgement timers

Triage Form Fields

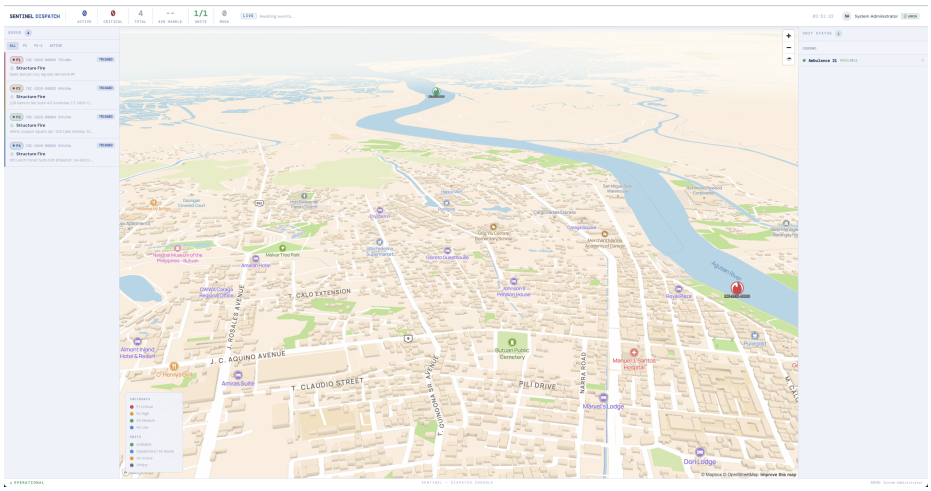
FIELD	TYPE	NOTES
Incident Type	Select	Full incident taxonomy with categories
Priority	P1-P4	Auto-suggested, operator confirms
Location	Text + Geocode	Address geocoded to coordinates + barangay
Caller Name	Text	For voice/walk-in reports
Caller Contact	Phone	Phone number for callback
Notes	Text	Operator notes and observations

DISPATCH LAYER

Dispatch Console

Real-Time Situational Awareness

The Dispatch Console provides a full 3D situational map powered by MapLibre GL JS. All incident markers and unit positions render as WebGL layers for maximum performance. Dispatchers assign units, track field operations, and manage the incident queue.



3D map with incident markers, unit tracking, and dispatch queue

Key Capabilities

- **3D map** — Real-time incident and unit markers with WebGL rendering
- **Priority queue** — Sortable, filterable queue with urgency indicators
- **Proximity assignment** — ETA calculation via road network routing
- **Ack timer** — 90-second auto-escalation if responder does not acknowledge
- **GPS tracking** — Live positions of all deployed units on map
- **Audio alerts** — Configurable alarm tones per priority level
- **Mutual aid** — Cross-agency coordination (BFP, PNP, DSWD, DOH)
- **Session metrics** — Incidents received, triaged, active, units deployed

Unit Status Indicators

STATUS	COLOR	DESCRIPTION
Available	Green	Ready for dispatch
En Route	Blue	Traveling to incident scene
On Scene	Amber	Arrived, operations active
Offline	Gray	Not logged in or unavailable

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RESPONDER LAYER

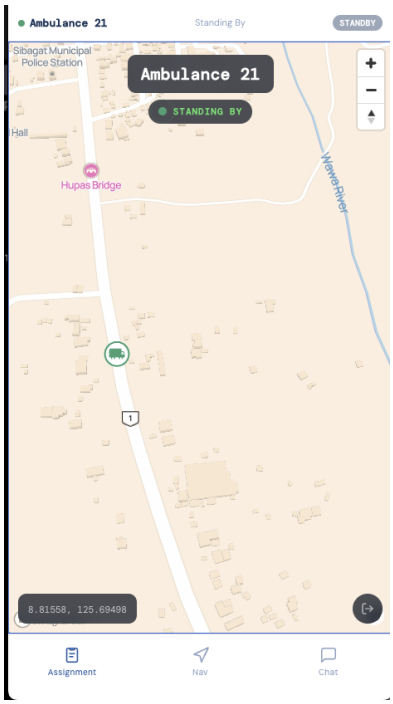
Responder PWA

Field Operations at Your Fingertips

The Responder application is a Progressive Web App for field teams. It provides assignment notifications, turn-by-turn navigation, scene documentation, and real-time communication with dispatch.

Status Workflow

STANDBY → **ACKNOWLEDGED** → **EN ROUTE** → **ON SCENE** → **RESOLVING** → **RESOLVED**



Key Capabilities

- **Push notifications** — Assignment alerts with 90-second ack timer
- **GPS tracking** — Real-time coordinate display and location sharing
- **Turn-by-turn navigation** — Route guidance to incident scene
- **Scene assessment** — Structured documentation and outcome recording
- **Resource requests** — Request additional support from dispatch
- **Closure summary** — Outcome (Transport, Treated, DOA, Stand Down)
- **Messaging** — Bi-directional real-time chat with dispatch

Status Definitions

STATUS	COLOR	DESCRIPTION
STANDBY	Steel	Available, no active assignment
ACKNOWLEDGED	Blue	Assignment received and confirmed
EN ROUTE	Blue	Traveling to incident scene
ON SCENE	Amber	Arrived, scene operations active
RESOLVING	Orange	Intervention complete, documenting
RESOLVED	Green	Incident closed, returning to standby

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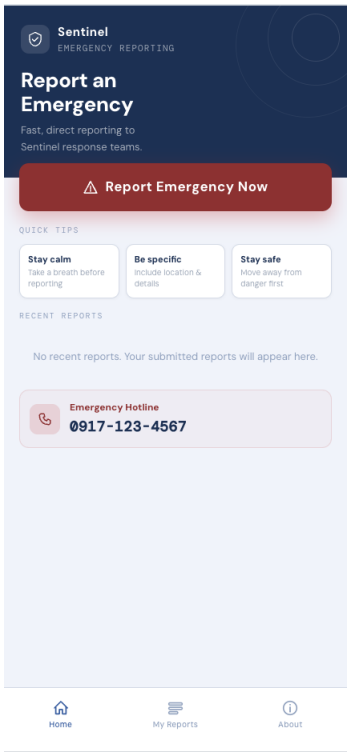
PUBLIC LAYER

Public Reporting

Empowering Citizens

The Sentinel reporting app gives citizens a direct line to emergency services. Anyone can report an emergency through their mobile browser. No app download required. Reports flow directly into the Intake Station for immediate triage.



Key Features

- **One-tap reporting** — Single button to initiate an emergency report
- **Location auto-detection** — GPS-based location with manual override
- **Report tracking** — Follow up on submitted reports
- **Emergency hotline** — Quick-dial to emergency services
- **Quick tips** — Safety guidance: stay calm, be specific, stay safe
- **No account required** — Accessible to all citizens without registration
- **Mobile-first** — Optimized for smartphone browsers

How It Works

→ **1. Report** — Citizen opens the web app and taps 'Report Emergency Now'

→ **2. Details** — Fills in incident type, location, and description

→ **3. Submit** — Report is sent directly to the Intake Station

→ **4. Triage** — Operator validates, classifies, and pushes to dispatch queue

→ **5. Response** — Nearest available unit is dispatched to the scene

The reporting app is a standalone Vue.js web application that communicates via REST API. It is deployed independently and can be accessed at any public URL, making it easy to distribute via QR codes, social media, or city notices.

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ACCESS CONTROL

User Roles & Permissions

Sentinel provides granular role-based access control. Each role aligns with specific operational responsibilities, ensuring users access only what they need.

PERMISSION	OPERATOR	DISPATCHER	RESPONDER	SUPERVISOR	ADMIN
Triage incidents	✓	—	—	✓	✓
Manual entry	✓	—	—	✓	✓
View dispatch map	—	✓	—	✓	✓
Create incidents	✓	✓	—	✓	✓
Assign units	—	✓	—	✓	✓
Update unit status	—	—	✓	✓	✓
View own assignment	✓	✓	✓	✓	✓
Close incident	—	—	✓	✓	✓
View analytics	—	—	—	✓	✓
Generate reports	—	—	—	✓	✓
Manage users	—	—	—	—	✓
System configuration	—	—	—	—	✓

Role Descriptions

- **Operator** — Monitors incoming channels across all five intake sources. Triages reports using structured forms, validates incident details, classifies priority, and pushes to the dispatch queue.
- **Dispatcher** — Manages the 3D dispatch map, assigns field units based on proximity and capability, tracks real-time field operations, and coordinates multi-unit responses.
- **Responder** — Receives assignment notifications on mobile, navigates to scenes, documents on-scene assessment and outcomes, requests additional resources, and files closure summaries.
- **Supervisor** — Oversees all layers with access to analytics dashboards, incident history, and performance metrics. Generates NDRRMC-compliant reports and conducts after-action reviews.
- **Administrator** — Full system access including user management, incident type configuration, system settings, barangay management, and all supervisor capabilities.

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ABOUT

Why Sentinel

Sentinel is purpose-built for Philippine local government emergency operations. Every component is open-source or locally owned, with zero vendor lock-in and zero recurring license fees.

Platform Advantages

- **Zero recurring license fees** — Fully owned codebase with no SaaS subscriptions or per-seat costs
- **Real-time operations** — WebSocket-powered live updates between all system layers
- **5-channel incident intake** — SMS, Mobile App, Voice/911, IoT Sensors, and Walk-in reporting
- **3D situational awareness** — WebGL-powered dispatch map with real-time unit tracking
- **Spatial intelligence** — Automatic geocoding and barangay boundary detection
- **Progressive Web App** — Installable mobile responder app with offline capability
- **Two-factor authentication** — Secure access for all operational roles
- **NDRRMC-compliant reporting** — Built-in report generation for national compliance
- **Multi-agency coordination** — Mutual aid protocol for BFP, PNP, DSWD, DOH
- **Full audit trail** — Timestamped incident timeline from report through resolution

Target Deployments

SECTOR	EXAMPLES
Disaster Risk Reduction	City/Municipal DRRMO offices, Provincial Emergency Operations Centers
Fire Services	Bureau of Fire Protection (BFP) stations
Law Enforcement	Philippine National Police (PNP) stations
Healthcare	Hospitals, medical facilities, ambulance services
Community	Barangay Emergency Response Teams (BERT)

SENTINEL INCIDENT RESPONSE MANAGEMENT SYSTEM

Developed by: HDSystem (HyperDrive System), Butuan City | **Client:** CDRRMO, Butuan City | **Region:** Caraga (XIII), Philippines

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